

Stochastic and Deterministic Modelling of Gene Expression using the Peccoud & Ycart Model

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1 Introduction

1.1 Biological background

Gene expression is one of the main biochemical processes required for any cellular function. It is part of the central dogma of molecular biology that genes encode mRNA, and that mRNA encodes protein. This process should be tightly controlled to produce the correct protein and sustain life. One mechanism of controlling gene expression is transcriptional induction. An inducible gene is usually not expressed (silent) until a regulatory factor is present and a regulatory interaction occurs, such as phosphorylation or ligand binding.

Gene expression is inherently stochastic due to the small number of molecules involved, especially at the single-cell level. As a result, the timing of reactions such as transcription and degradation is random, leading to variability in protein levels even between identical cells.

1.2 Model description

In this project, the gene expression model proposed by Peccoud and Ycart (1995) is replicated in COPASI. The Peccoud & Ycart model is a stochastic model of gene expression that models gene product synthesis within a single cell as a continuous-time Markov process. In this project, a simplified model is used, in which mRNA dynamics are not explicitly included.

According to this model, the gene can be in an active (ON, 1) or inactive (OFF, 0) state. The transition rates between these states are λ for $\text{OFF} \rightarrow \text{ON}$ and μ for $\text{ON} \rightarrow \text{OFF}$, making this a continuous-time two-state Markov process.

When the gene is ON, its product is produced at a rate ν . Proteins are degraded at a rate δ . The system tracks how many protein molecules are present and whether the gene is ON or OFF.

1.3 State space and dynamics

The state of the system is of the form (i, n) , where $i \in \{0, 1\}$: gene ON/OFF and $n \in \mathbb{N}$.

Possible transitions:

Gene switches:

- $(0, n) \rightarrow (1, n)$ at rate λ
- $(1, n) \rightarrow (0, n)$ at rate μ

Protein dynamics:

- Production (if gene ON):
 - $(1, n) \rightarrow (1, n + 1)$ at rate ν
- Degradation:
 - $(i, n) \rightarrow (i, n - 1)$ at rate $n\delta$

This formulation defines a continuous-time Markov chain over the state space of gene states and protein counts.

In this setup, protein production occurs in bursts because it is only possible when the gene is in the ON state. Since the gene randomly switches between ON and OFF states, production happens intermittently, leading to sudden increases in protein numbers. In contrast, degradation occurs continuously, with each protein molecule independently degraded at a constant rate over time, resulting in a gradual decrease proportional to the number of proteins present.

2 Model description in COPASI

2.1 Initial values of metabolites

Under the assumption that the gene is initially inactive (OFF) and there are no protein copies:

- $G_{off}(0) = 1$
- $G_{on}(0) = 0$
- $P(0) = 0$

2.2 Model reactions



With the conservation rule: $G_{off} + G_{on} = 1$. That means the gene exists in exactly one copy and is either OFF or ON.

2.3 Model parameters

The original article defines the model symbolically in terms of the parameters λ , μ , ν , and δ , but does not prescribe a unique numerical parameter set. Therefore, for simulation in COPASI, parameter values were chosen to produce representative system dynamics and to illustrate the stochastic behaviour of the model. The selected parameter values are consistent with typical ranges used in simplified stochastic gene expression models.

Parameter	Meaning	Unit	Value used in simulations
λ	Rate of transition from inactive to active gene state	time^{-1}	$\lambda = 0.05$
μ	Rate of transition from active to inactive gene state	time^{-1}	$\mu = 0.1$
ν	Protein synthesis rate in the active gene state	$\text{molecule} \cdot \text{time}^{-1}$	$\nu = 1.0$
δ	Protein degradation rate per protein molecule	time^{-1}	$\delta = 0.02$

Table 1: Model parameters.

2.4 Model description in COPASI

Deterministic simulation was performed using the LSODA integrator (Time Course task) with duration 200 time units and 1000 output intervals. The protein copy number and gene activity (G_on) were recorded over time.

Stochastic simulation was performed using the Direct Method (Gillespie algorithm), implemented via the Parameter Scan task with 10 repeated runs of the Time Course task, each starting from the same

initial conditions. This produces 10 independent stochastic trajectories representing 10 virtual cells. Duration and intervals were identical to the deterministic case (200 time units, 1000 intervals).

Both simulations start from the same initial state: $G_{off} = 1$, $G_{on} = 0$, $Prot = 0$, consistent with the biological assumption in Peccoud & Ycart (1995) that the gene is initially inactive and no protein is present.

2.5 Running simulations in COPASI

To reproduce the simulations, open the provided COPASI file `project.cps` and proceed as follows. Set "Particle Numbers" instead of "Concentrations" in the top menu. For the deterministic simulation, select the Time Course task, ensure the method is set to Deterministic (LSODA), and click Run. For the stochastic simulation, set the Time Course method to Stochastic (Direct method), then select the Parameter Scan task and click Run. In both cases the plot "Protein dynamics" will be generated automatically.

3 Results

As Figure 1 shows, deterministic simulation produces a smooth sigmoid-like increase in protein copy-number. Protein copy-number curve converges to a steady-state of approximately 16.7 molecules. This number of molecules is consistent with analytical prediction: $E = \frac{\lambda\nu}{(\lambda+\mu)\delta} = \frac{0.05 \times 1.0}{(0.15) \times 0.02} = 16.7$. The gene activity G_{on} stabilises at 0.33, equal to $\frac{\lambda}{(\lambda+\mu)}$, which confirms that gene spends around 33% of time in the active state.

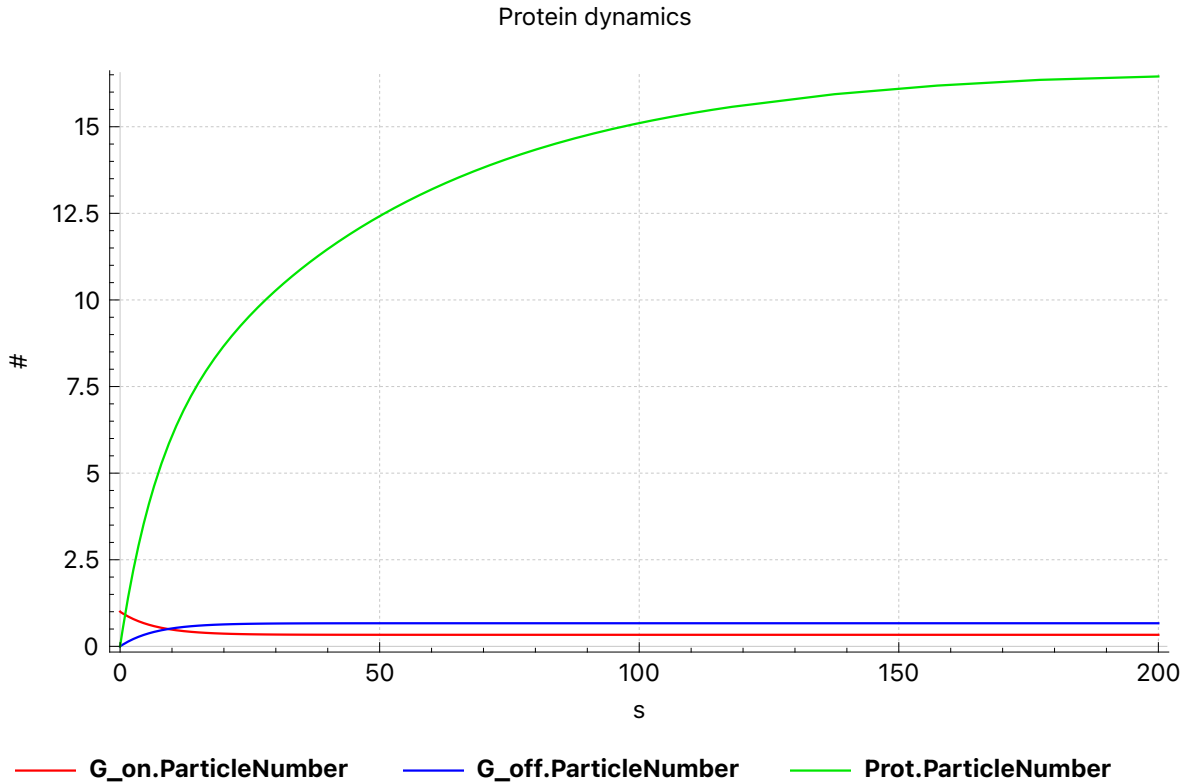


Figure 1: Deterministic simulation. Protein copy number (green) rises to steady state 16.7. G_{on} (blue) and G_{off} (red) stabilise at 0.33 and 0.67 respectively.

Figure 2 shows a single representative trajectory in detail, illustrating the burst-like production pattern that underlies the variability seen across all trajectories in Figure 3. The sudden increase in the number of protein molecules corresponds to a burst of protein production when the gene switches ON. The burst is followed by a gradual exponential decay during OFF periods. The underlying binary gene

switching process is visible: G_on alternates stochastically between 0 and 1, with the average dwell time in the ON state of $\frac{1}{\mu} = 10$ time units and in the OFF state of $\frac{1}{\lambda} = 20$ time units.

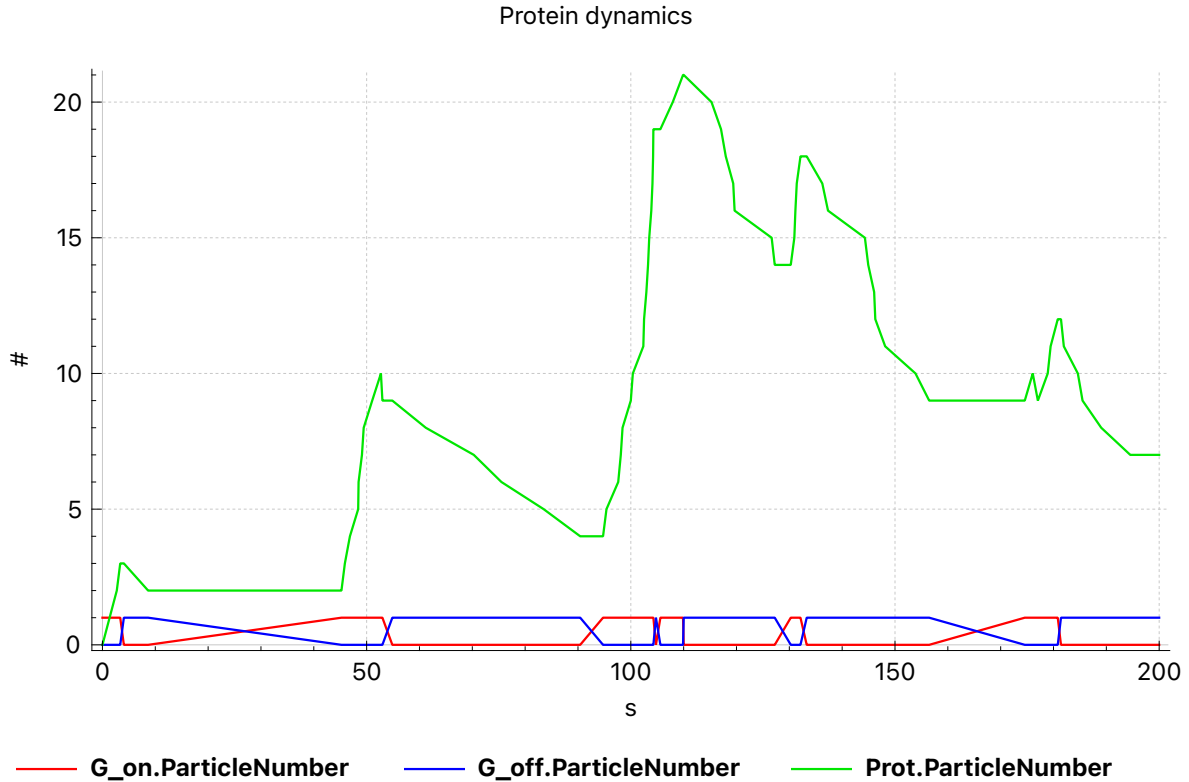


Figure 2: Stochastic simulation single trajectory. Protein copy number (green). G_on (blue) and G_off (red) oscillate between two states: 0 and 1.

Figure 3 shows stochastic simulation with 10 independent trajectories, each representing a single cell. All trajectories are around the deterministic mean of 16.7 molecules but also show substantial variability.

Stochastic trajectories exported from the COPASI software were analysed using a Python script (`data_analysis.ipynb`). Each trajectory was interpolated onto a uniform time grid of 1000 points to enable pointwise computation of the mean and standard deviation across the 10 runs. Steady-state statistics were computed over the final 10% of the simulation ($t = 180\text{--}200$). Figure 3 shows the resulting plot with all trajectories and their mean and variance. Table 2 presents summary statistics of protein molecule number at steady-state across 10 trajectories.

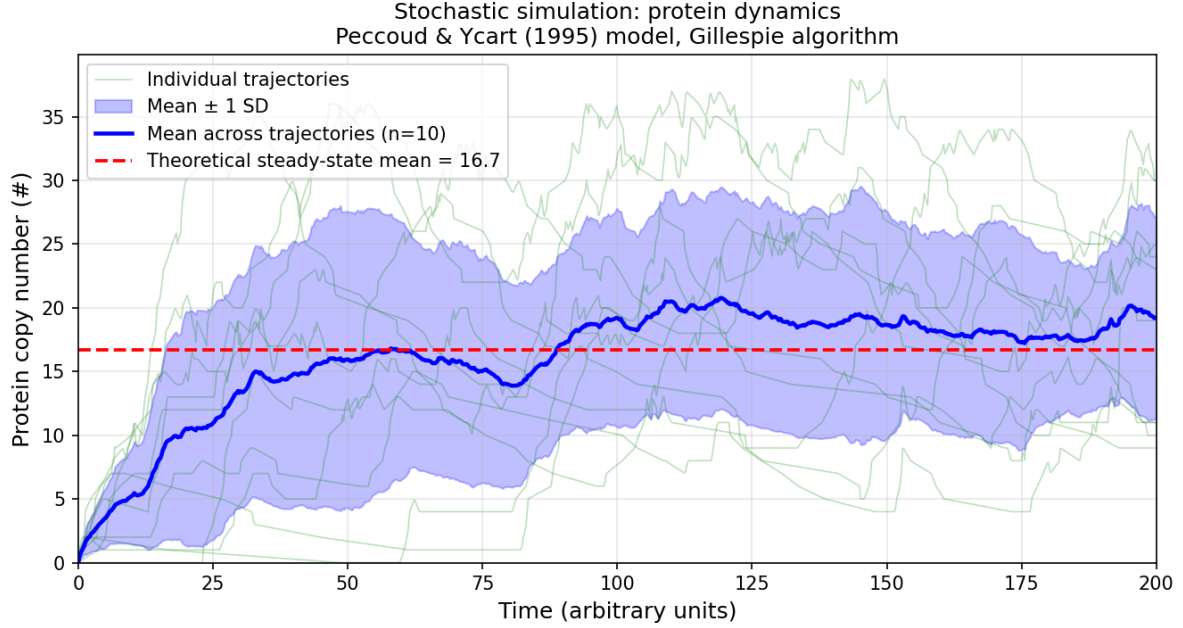


Figure 3: Stochastic simulation of protein dynamics using the Peccoud & Ycart (1995) model. Ten independent trajectories (green) were generated using the Gillespie Direct Method in COPASI, each representing a single cell starting from the same initial conditions ($G_{\text{off}} = 1$, $G_{\text{on}} = 0$, $\text{Prot} = 0$). The bold blue line shows the mean protein copy number across all 10 trajectories, with the shaded blue region indicating ± 1 standard deviation. The dashed red line marks the analytical steady-state prediction of 16.7 molecules. The simulated mean at steady state ($t = 180\text{--}200$) was 18.5 ± 6.0 molecules, consistent with the theoretical value given the small sample size of 10 trajectories.

Table 2: Summary statistics of protein copy number at steady state ($t = 180\text{--}200$) across 10 stochastic trajectories.

Statistic	Value	Notes
Number of trajectories	10	Independent Gillespie runs
Theoretical steady-state mean	16.67 molecules	$\mathbb{E} = \frac{\lambda\nu}{(\lambda+\mu)\delta}$
Simulated mean ($t = 180\text{--}200$)	18.50 molecules	Average across 10 trajectories
Standard deviation	6.03 molecules	Cell-to-cell variability
Variance	36.42 molecules ²	
Min trajectory mean	9.76 molecules	Most protein-poor cell
Max trajectory mean	25.70 molecules	Most protein-rich cell

4 Discussions

The deterministic and stochastic simulations both predict a steady-state protein level of approximately 16.7 molecules, as given by the analytical calculation (see 3). However, the two simulations differ in what they reveal about the system’s behaviour.

The deterministic simulation produces a single smooth trajectory that monotonically approaches the steady state. This corresponds to the notion that molecular species are present in large and continuous quantities. In this case, every simulation starting from the same initial conditions produces exactly the same result, and there is no variability between cells.

The stochastic simulation reveals a different behaviour. The 10 independent trajectories differ substantially from one another, with steady-state protein counts ranging from 9.8 to 25.7 molecules (Table 2). This variability arises from two sources of intrinsic noise: the random switching of the gene between ON and OFF states, and the probabilistic nature of individual protein production and degradation events.

This result directly illustrates the central argument of Peccoud & Ycart (1995): that stochastic gene expression produces significant cell-to-cell variability in protein levels even among genetically identical cells in identical environments. The deterministic ODE model treats the gene as having a continuous fractional activity of $\frac{\lambda}{\lambda+\mu} = 0.33$, which is biologically unrealistic at the single-cell level, since single gene copy is either ON or OFF, never partially active.

The stochastic mean across 10 trajectories (18.5 molecules) is slightly higher than the theoretical prediction (16.7 molecules). This deviation is expected given the small sample size of only 10 trajectories and would decrease with a larger number of runs. Despite this, the simulated mean is within one standard deviation of the theoretical value, confirming that the implementation is correct.

In conclusion, the deterministic model is appropriate for predicting population-average protein levels, while the stochastic model is necessary to capture the single-cell behaviour and intrinsic noise that characterises gene expression in vivo. This shows that even a simple four-reaction model is sufficient to capture the stochastic variability observed in real gene expression experiments.

References

Peccoud, J. and Ycart, B. (1995). Markovian modeling of gene-product synthesis. *Theoretical Population Biology*, 48(2):222–234.